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SURVEY

Plant Disease Detection Using Self-Supervised Learning: A Systematic Review

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ABSTRACT Agriculture has a crucial role in both the economy and food supply. However, the frequent occurrence of plant diseases can have a significant negative influence on the production of foods. Timely detection of a disease is important for its effective management, e.g., targeted use of pesticides. Conventional Plant Disease Detection (PDD) methods are manual, slow, tedious, and prone to errors, thereby increasing the risk of significant yield losses. Recently, various data-driven approaches, including Deep Learning and Computer Vision-based approaches, have been explored for PDD. However, the scarcity of large and annotated datasets and the limited scalability of these approaches have prompted researchers to turn to the Self-Supervised Learning (SSL) approach. In this review, we provide the very first conceptual grounding for the SSL approach in PDD. By reviewing a large body of recent related works in the literature, we thoroughly analyse and categorise them into generative, predictive, contrastive and hybrid SSL models. Moreover, this review analyses various recent datasets and performance metrics used in these models. Finally, we explain the research challenges and key research directions aimed at advancing PDD through the SSL approach.

INDEX TERMS Digital agriculture, self-supervised learning, plant disease detection, machine learning, deep learning, image processing.

I. INTRODUCTION

Agriculture stands as one of the oldest and most important industries in the world, and the demand for a proper supply of food is increasing as the population grows [1]. Ensuring the production of high-quality plants is imperative, given their susceptibility to various biotic stresses [2]. Frequent and long-lasting diseases to agricultural plants can cause economic damage to a nation as well as bring famine to its population [3]. These diseases are difficult to monitor at a large scale, making them difficult to control and eradicate. More importantly, misdiagnosis of disease can lead to misuse of chemicals, learning to the emergence of resistant pathogen strains, increased input cost, and more outbreaks with significant economic loss and environmental impacts. Thus, identifying a disease correctly in an early stage is a crucial step for efficient and effective disease

management and improvement of plant yield in terms of both quality and quantity in modern agriculture. While the human visual system excels at pattern recognition, it introduces subjectivity, optical illusions and errors. Moreover, disease diagnosis based on human inspection is time-consuming and expensive. To address these challenges, various efforts have been made to prevent plant disease in the past decade [4], [5]. In recent years, automated techniques such as Artificial Intelligence (AI) hold promise for efficient, accurate, and rapid disease detection in smart farming and precision agriculture [6], [7], [8].

In fact, AI is being used on a large scale in smart agricultural systems across five key sectors: plant yield prediction, stress detection, weed and pest detection, smart farming, and Plant Disease Detection (PDD) [9]. In plant yield prediction, detection techniques such as YOLO-v2 [10], YOLO-v3 [11], CNN [12] and RCN [13] are commonly utilised to identify and quantify plant production within specific areas. Stress detection in plants is facilitated by a range

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of approaches, including parametric [14], nonlinear [15], and radiative transfer [16] methods, aimed at recognising indicators of plant stress. In weed and pest detection, different convolution neural networks [17], [18], [19] have been employed to identify these threats among plants. Similarly, in smart farming applications, models such as Conv3D [20], ConvLSTM [21], and CNN-LSTM [22] are utilised to predict soil temperature, a crucial factor in agriculture management. The good quality of plant production is intrinsically linked to plant health, making automated PDD a vital application of AI [23], [24].

A. PLANT DISEASES AND THEIR IMPACT

Generally, there are many types of plant diseases that depend on many factors. We can divide plant diseases into three categories: primary reason, spread, and symptoms. Figure 1 shows the sample images of different plant diseases. Plant diseases are caused primarily by fungal, viral and bacterial organisms. While some diseases (e.g., orange rust in sugarcane) can be identified by visual symptoms like blight, rust and rot, many others (e.g., ratoon stunting disease in sugarcane) have no visual symptoms. They can be spread in the environment via soil, water, air and seeds.

If these diseases are not detected and managed in a timely manner, they can cost us significantly. For example, the severe outbreak of wheat rust in Australia in 1973 reduced Australian wheat production by 25%-35% [25]. The top plant disease, *Xylella fastidiosa*, threatens 350 native plant species in Australia and often causes leaf scorch, bacterial leaf scorch, olive leaf, etc. It is reported that the economic impacts of *X.fastidiosa* on the grape wine industry ranged from \$2.8 billion to \$7.9 billion at 2014-15 prices [26]. Rice blast, a severe fungal (*Oryza sativa* L.) disease affecting rice, poses a significant threat to global food security [27]. This disease accounts for annual yield losses ranging from 10% to 30%. Under optimal conditions, rice blast can destroy entire rice plants within 15 to 20 days, potentially resulting in total yield losses of up to 100% [28].

B. CHALLENGES IN DETECTION

Early and accurate detection of plant diseases holds the key to minimising reliance on unsafe chemicals in plant cultivation and protection, which is averting potentially serious consequences for plant quality, quantity, and productivity. Timely identification of symptoms at their incipient stages on plant leaves can significantly reduce the labour-intensive monitoring required across extensive agricultural holdings. Despite the prevailing reliance on manual detection methods due to the inherent challenges of early disease identification, the economic significance of agriculture underscores the urgency of advancing disease identification and classification practices within the industry [29], [30], [31].

Although modern AI-based methods have the promise to increase the efficiency of PDD, the great variance in symptoms due to the age of infected tissues, genetic

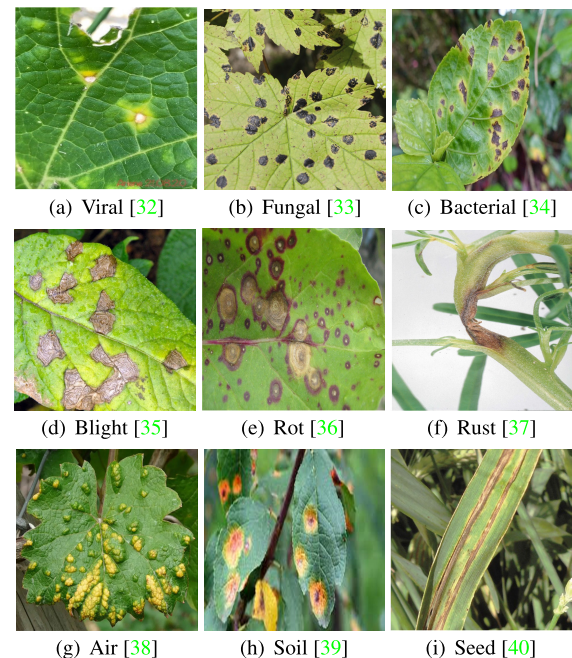


FIGURE 1. Examples of plant diseases: (a-c) caused by organisms, (d-f) visible symptoms, and (g-i) spread via media.

variations, and label scarcity on the datasets decrease the accuracy of detection. The label scarcity on a dataset especially weakens the learning ability of the AI models. It is very difficult to create high-quality and diverse datasets that greatly enhance the performance of AI models. The data should be collected in such a way that it includes different types of environmental conditions, species, and parts of the plant clearly and accurately. Collecting this data is expensive, time-consuming and requires a lot of manpower and infrastructure to put it to good use in the agricultural world. An even bigger challenge is to develop an AI model that has low computational complexity and generalisability to unseen datasets. PDD also faces other challenges, such as class imbalance, feature extraction and compatibility.

C. MOTIVATION FOR SELF-SUPERVISED LEARNING

Deep Neural Networks (DNNs) have demonstrated remarkable efficacy across diverse Machine Learning (ML) tasks, particularly in supervised learning within computer vision and Natural Language Processing (NLP) domains [41], [42], [43], [44]. However, supervised learning encounters challenges in PDD, requiring extensive labelled datasets that are costly and time-intensive to annotate. Additionally, managing large volumes of data on specific plant diseases poses logistical challenges.

In response to these challenges, self-supervised learning (SSL) has emerged as a promising alternative, particularly in scenarios with limited or unstructured data availability [45]. SSL differs from supervised learning in that it generates its own labels from input data, aiming to maximise learning efficiency from vast amounts of unlabeled data. SSL has

garnered significant attention owing to its data efficiency and generalisation capabilities.

Recent advancements in Deep Learning (DL) have revolutionised disease detection and classification, leveraging image-based analysis to categorise plant diseases [46], [47], [48], [49], [50]. Self-supervised methods further enrich the diagnostic toolkit available to agricultural stakeholders [4], [51], [52]. This approach facilitates the maximisation of unlabelled data resources, thereby optimising the model's learning efficiency for PDD [53].

D. PAPER CONTRIBUTIONS

In the domain of using AI in PDD, although there is no systematic review using SSL models, there are many recent survey articles exploiting other AI models, including transfer learning. Many of these surveys present reviews on using AI in the general agricultural field [9], [54] and other reviews specifically focus on AI and PDD [55], [56].

Rani et al. [54] introduce a systematic survey, delineating multiple facets of recent advancements of AI techniques in the agricultural field. Li et al. [57] discuss label-efficient learning in agriculture, including PDD. Whereas the studies [9], [58] explain the application of DL techniques in the agricultural field. Shafik et al. [59] provides a comprehensive analysis of strategies for detecting plant diseases using image-based techniques. It covers both the localisation and classification aspects of disease detection. The paper [56] provides a comprehensive overview of vision-based ML and transfer learning methodologies utilised for PDD. Lastly, Varshney et al. [55] present an extensive review encompassing various research endeavours focused on PDD, employing both state-of-the-art handcrafted features and DL methodologies.

Table 1 summarises the contributions of current review articles focusing on AI in PDD. It shows there is no detailed discussion focusing only on the application of SSL models in PDD. The majority of review articles within the domain of PDD have extensively covered topics such as DL, ML, transfer learning, and conventional methodologies, meticulously examining their respective advantages and limitations. However, it is noteworthy that only a limited number of papers have provided cursory discussions on SSL models, with a singular paper delving into the application of such models specifically within PDD. To date, no review paper has been dedicated solely to the exploration of self-supervised models in the context of PDD. Consequently, there is sufficient scope for a systematic review article to discuss self-supervised models used only in PDD.

The principal aim of this review paper is to comprehensively examine the influence, results, potential challenges, and datasets employed by SSL methodologies in the realm of PDD. By thoroughly scrutinising these aspects, the intention is to lay the groundwork for the future development of intelligent SSL techniques tailored specifically for PDD applications. The key contributions to the review are as follows.

TABLE 1. Recent review articles for plant disease detection.

Topic	Article reference						
	[54]	[57]	[9]	[58]	[59]	[56]	[60]
Systematic review	✓	✓			✓	✓	✓
Transfer learning						✓	✓
Machine and deep learning	✓	✓	✓	✓		✓	✓
Brief discussion on SSL	✓	✓	✓		✓		✓
Dataset used in SSL							✓
Categorical discussion of SSL							✓
Challenges of SSL							✓

Note: SSL = Self-Supervised Learning.

- To the best of our knowledge, this is the first systematic review that solely discusses SSL methods in PDD based on four categories: generative, predictive, contrastive and hybrid learning.
- The review discusses the datasets and performance metrics used in SSL for PDD.
- The review also explains the research challenges of SSL in PDD.

E. ORGANIZATION

The remainder of the manuscript is organized as follows: Section II provides the background on PDD and SSL. Section III details the methodology used for the paper selection process. Section IV discusses the answers to the research RQs and highlights the research challenges encountered. Finally, Section V presents the conclusions and outlines future research directions in the application of SSL for plant disease detection.

II. BACKGROUND STUDY

In this section, we explore various traditional image processing, ML, and deep learning (DL) approaches applied to PDD. We also provide an overview of different categories of self-supervised learning (SSL), as this manuscript focuses only on SSL approaches to PDD.

A. PLANT DISEASE DETECTION BACKGROUND

AI has played a very important role in the agricultural field over the past years, especially in the context of PDD. Initially, PDD was investigated through traditional image processing approaches [61], [62], [63], [64]. Later, due to the development of ML algorithms [5], [6], [65] in other domains, its use in PDD also increased. Different types of pre-processing approaches like Gaussian function [66], background removal [67], colour constancy [68], resize and data augmentation [68], [69], [70] were applied to the input images. Feature extraction approaches such as colour transformation [5], [63], [68], [71], [72], local binary patterns [61], [62], [65], [73], colour features [61], [74], [75], and histograms [64], [66], [73], [74], [76], [77] were often used in different classifiers like support vector machine [6], [76], [78], [79], rule-based classifier [7], [80], k-nearest neighbours [5], [65], [76], [81] to classify or detect plant diseases.

Following the introduction of DL, researchers achieved remarkable success by applying various DL algorithms to the

domain of PDD. These algorithms, particularly convolutional neural networks (CNNs), excel at processing vast amounts of data by automatically learning complex and abstract features through multiple layers of convolutions. Unlike traditional methods that rely on manual feature extraction, CNNs can identify intricate patterns in images, such as subtle variations in leaf colour, texture, and shape, which are indicative of specific diseases. The architecture of these models, including well-known variants like AlexNet, VGGNet, ResNet, and EfficientNet, enables them to capture hierarchical features. Initial layers detect simple edges and textures, while deeper layers identify more sophisticated representations. This hierarchical learning process allows DL models to achieve higher accuracy and robustness in classifying and detecting plant diseases.

Specifically, the researchers applied VGG16 [8], [82], [83], [84], AlexNet [8], [85], [86], convolutional encoder [87], ResNet50 [48], [88], [89], SegNet [90], few-shot [91], dense CNN [92] and generative [93], [94] DL network for PDD. Furthermore, some studies have attempted to detect plant diseases using transfer learning [95], [96], [97], attention-based CNN [98], [99] and lightweight CNN [100], [101], [102].

Researchers are also using SSL techniques [4], [103], which can detect or classify plant disease without the labelled dataset. Additionally, advanced methods like data augmentation and synthetic data generation using generative adversarial networks (GANs) address the challenge of limited training data, further improving the models' generalisation capability.

B. SELF-SUPERVISED LEARNING AND ITS CATEGORIES

The issue of inadequately labelled datasets in PDD poses a significant difficulty as conventional ML and DL models largely depend on extensive volumes of annotated data to acquire knowledge and make precise predictions. Assigning labels to such data necessitates specialised expertise and entails a significant investment of time and resources. SSL is a potential approach to address this problem since it allows models to acquire valuable representations from data without relying on labels. SSL techniques let models acquire an implicit understanding of the data's structure and properties, equipping them to carry out tasks such as classification and detection with little or no dependence on labelled datasets.

The SSL consists of two tasks such as pretext and downstream tasks. The actual self-supervised approach happens in the pretext task, where the model is trained in a supervised manner to have labels from the unlabelled data. Hence, the model acquires significant information from the input data. Subsequently, the acquired information is transferred into the downstream task to complete the required objectives. Figure 2 illustrates the primary working process of a self-supervised approach. This method can be very beneficial in the field of plant disease identification, as there is often a scarcity of datasets that have been annotated by experts. The application

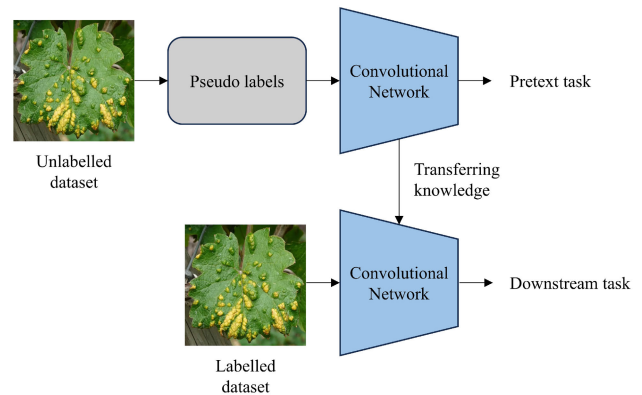


FIGURE 2. A sample primary working process of a self-supervised approach.

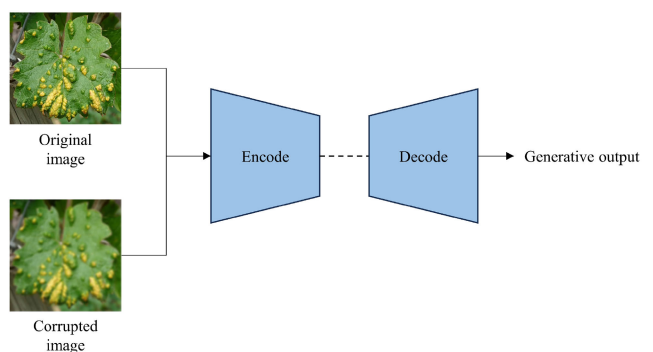


FIGURE 3. A sample workflow of a generative self-supervised learning approach.

of SSL in PDD can be classified into four primary categories: Generative, Predictive, Contrastive, and Hybrid.

1) GENERATIVE SELF-SUPERVISED LEARNING

Generative SSL approaches [23], [51], [104] are trained to create new data samples that closely resemble the original data. Within the field of PDD, these models possess the ability to acquire knowledge about the distribution of images indicating healthy and diseased plants. Consequently, they are capable of generating novel images representing both healthy and afflicted plants. This facilitates comprehension of the fundamental patterns of diseases and may also be helpful in anomaly identification by spotting images that deviate from the learned distribution. A sample workflow of the generative approach is depicted in Figure 3, which utilised the encode-decode approach to generate the output.

The generative network can be an autoencoder and a Generative Adversarial Network (GAN). An autoencoder comprises an encoder, which compresses data into a latent space, and a decoder, which reconstructs the original data from this compressed representation. It evolves several adaptations to enhance its capability and efficiency. Early autoencoders [105] use a bottleneck structure to compress inputs into a lower-dimensional latent space, similar to a saliency map [106], primarily for dimensionality reduction. Sparse autoencoders, on the other hand, incorporate sparsity

constraints to allow the latent space to be larger than the input dimension. Denoising autoencoders (DAEs) [107] improve robustness by reconstructing data from its noise-corrupted versions, which helps the network to focus on the most significant features. Variational autoencoders (VAEs) [108] add a probabilistic layer by encoding inputs into Gaussian distributions, optimising not only the reconstruction loss but also minimising the divergence to enhance the generalisability of the learned features with normalisation flows [109]. Most recently, masked autoencoders (MAEs) [110] have been developed for computer vision to process the remaining visible parts with the encoder and task the decoder with reconstructing the missing parts.

Generative Adversarial Networks (GANs) [111] operate on an adversarial training framework where a generator (G) creates synthetic samples from random latent vectors, and a discriminator (D) attempts to distinguish these from real data. Adversarial Autoencoders (AAEs) [112] integrate autoencoders by using the encoder to generate latent vectors for real samples, enhancing the training process. Cascade Wavelet Attention Network (CWAN) [23] is also used for synthesising and segmenting images.

2) PREDICTIVE SELF-SUPERVISED LEARNING

Predictive SSL methods [4], [113] are primarily concerned with forecasting a portion of the input data based on other portions using the pretext information. For instance, when provided with a portion of a plant image, the model may generate a prediction for another corresponding portion. During this phase, the model acquires the ability to comprehend the arrangement and circumstances of the images, which is essential for recognising distinct disease patterns in plants. A sample workflow of the predictive approach is depicted in Figure 4.

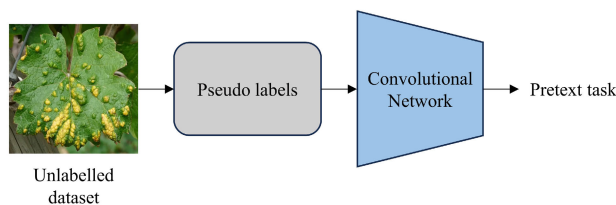


FIGURE 4. A sample workflow of a predictive self-supervised learning approach.

Spatial [114], spectral [115], and temporal [116] contexts play an important role in predictive self-supervised models. Spatial context is about understanding the layout and relationships between elements in images or spatial data, which aids the model in completing or segmenting images. Spatial predictive approaches can be applied for a relative position [117], jigsaw [118], rotation [119], and inpainting [120]. For instance, the Location Feedback Classification Network (LFC-Net) [113] was applied to identify the relative spatial information for identifying plant diseases. Spectral context refers to the analysis of data

across several wavelengths or frequencies, such as those seen in satellite images or sounds. In this context, the model is trained to improve or understand distinct spectrum characteristics. For instance, spectral predictive approaches can be used for colourisation [121] to predict the colourful images from grey images. Temporal context describes the ability to recognise and analyse patterns that occur across time. This is particularly important when dealing with videos or time-series data. By comprehending temporal context, a model can make predictions about future occurrences or fill in missing sequences of data [116].

3) CONTRASTIVE SELF-SUPERVISED LEARNING

Contrastive SSL models [45], [51], [122] are predicated on the concept of acquiring knowledge through the process of comparison. The goal of training models is to identify pairs of data samples that are similar (positive) and pairs that are different (negative). In PDD, this process includes distinguishing between images of healthy plants and those displaying disease symptoms. By doing so, the system can acquire knowledge of unique characteristics without relying on explicit labels. A sample workflow of the contrastive approach is depicted in Figure 5.

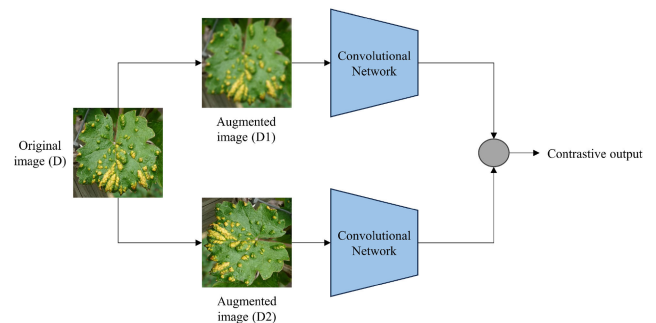


FIGURE 5. A sample workflow of a contrastive self-supervised learning approach.

Contrastive SSL models utilise a range of approaches, including negative sampling [123], clustering [124], knowledge distillation [125], and redundancy reduction [126]. Negative sampling aids in the discrimination of similar and dissimilar samples, so efficiently instructing the model to identify and discern crucial characteristics. Vision Transformer (ViT) [127], Self-supervised Contrastive learning method (SimCLR) [128], and Triplet loss [129] can be used as Negative sampling. Clustering is a pretext task in which the model clusters together data points that are similar, allowing it to learn distinctive characteristics that can identify these groupings. Cross-iterative kernel-based Image Clustering System (CIKICS) [122], Contrastive Learning for Leaf Disease Identification (CLA) [53], Self-Labeling (SeLa) [130] and Prototypical Contrastive Learning (PCL) [131] are some examples of the clustering process. Knowledge distillation is the process of transferring knowledge from a larger and more intricate model to a smaller one. The

objective is for the smaller model to imitate the performance of the larger model while utilising fewer computer resources. Redundancy reduction aims to minimise the encoding of redundant information, allowing the model to capture only the most crucial feature. Simple Siamese (SimSiam) [132] and self-distillation with no labels (DINO) [133] can be applied as knowledge distillation SSL approaches.

4) HYBRID SELF-SUPERVISED LEARNING

Hybrid techniques [134], [135] integrate components of generative, predictive, and contrastive learning to exploit the advantages of each. Hybrid models can enhance their capacity to identify and categorise plant diseases in diverse settings by employing a multifaceted strategy, leading to a more thorough comprehension of the data. A sample workflow of the hybrid approach is depicted in Figure 6, where a convolutional autoencoder was used with an additional convolutional network.

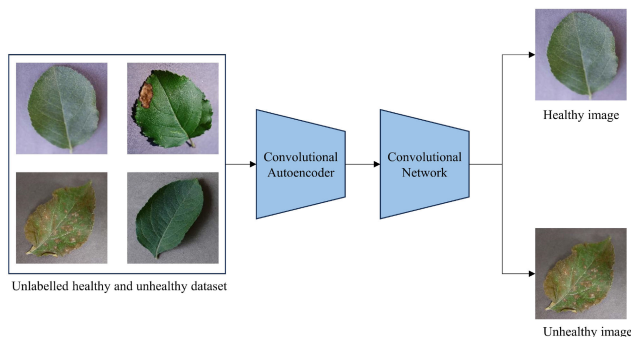


FIGURE 6. A sample workflow of a hybrid self-supervised learning approach.

III. SYSTEMATIC REVIEW METHODOLOGY

The current review adhered to the guidelines set forth by the Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA) [136]. Initially, the authors independently screened the titles and abstracts of the publications retrieved via database search. Subsequently, a comprehensive examination of all pertinent studies was conducted by a thorough review of their full-text content. This approach utilized the following procedures:

A. IDENTIFYING RESEARCH QUESTIONS

For the purpose of conducting this systematic review, the following Research Questions (RQs) were formulated in the context of PDD using SSL models:

RQ1: What are the SSL algorithms utilised?

RQ2: Which datasets have been employed?

RQ3: Which performance metrics have been used to evaluate the effectiveness of the models?

The response to this question will contribute to enhancing SSL algorithms for PDD and pinpointing the research gaps in the existing methodology.

B. DATABASE

Extensive literature searches were conducted across five reputable repositories, including Scopus, IEEE Xplore and ScienceDirect, to identify relevant sources for this study.

C. SEARCH STRATEGY

To comprehensively explore self-supervised approaches for PDD, the following keywords were considered: self-supervised, plant disease, and leaf disease. The search query string employed was: (“Self-supervised” AND (“plant disease” OR “Leaf disease”)). The search scope was restricted to articles published between 2020 and 2024, specifically focusing on SSL in PDD.

D. CRITERIA FOR ELIGIBILITY

The eligibility criteria, which include the inclusion and exclusion criteria, are established to identify suitable manuscripts for this review process. The inclusion criteria are: 1) the language is English; 2) the manuscript that discussed PDD and SSL; 3) full-length journals and conference manuscripts; and 4) the manuscript that was published between 2020 and 2024. Again, the exclusion criteria are: 1) the language is not English; 2) the manuscript that discussed PDD except SSL models; 3) not peer-reviewed, poster, preprint and short articles; and 4) the manuscript with irrelevant facts. In the context of PRISMA, including manual searches to capture relevant studies not found through database searches is acceptable and sometimes necessary [137], especially when recent papers (e.g., conference papers and technical reports) are not indexed in standard databases. This ‘Manual Search’ approach has already been applied in several PRISMA-based reviews in recent literatures [138] and [139] to include studies that meet the inclusion criteria. Hence a manual search strategy is also applied in this study to include the relevant manuscript that is not found in the search database.

E. ARTICLE SELECTION AND EXTRACTION PROCESS

Following an initial search, a total of 422 publications were identified across the specified five databases. Prior to screening, 109 duplicate articles and 23 review articles were excluded. Subsequently, 290 articles underwent screening based on title and abstract, resulting in the removal of 189 articles during title screening and an additional 83 articles during abstract screening. Based on predefined eligibility criteria, 18 articles were deemed eligible for review, with one additional article identified through a manual search. Ultimately, a comprehensive review was conducted on a total of 19 articles for the systematic analysis. The procedural steps of the PRISMA method are depicted in Figure 7.

IV. RESULT AND DISCUSSION

In this section, we address the RQs based on the findings from the PRISMA framework. We provide a detailed analysis

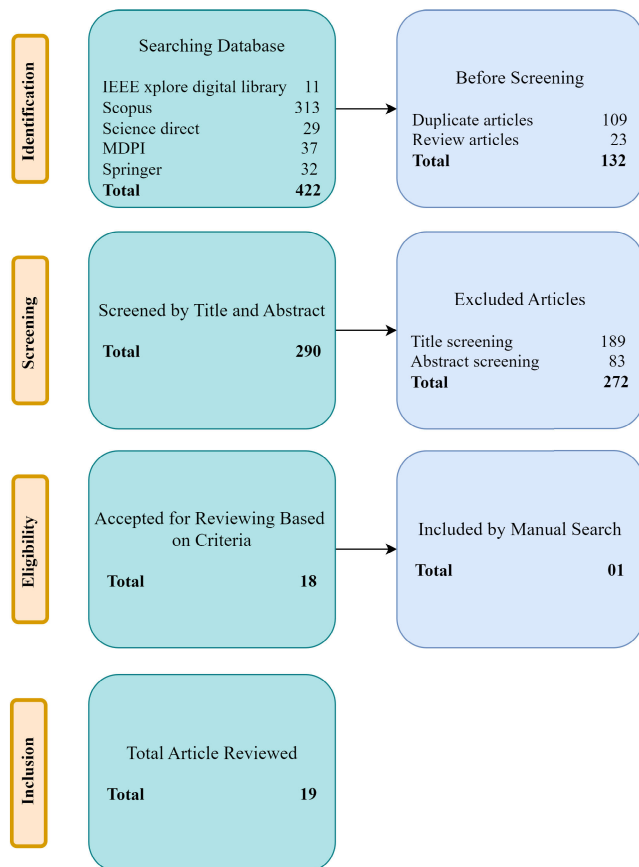


FIGURE 7. PRISMA flow diagram for the manuscript selection process.

of the selected studies and their relevance to each RQ. The section also discusses the key research challenges identified during the review. Additionally, we highlight gaps in the literature and suggest future research directions to advance this field.

A. RESULTS

1) RQ1: WHAT ARE THE SSL ALGORITHMS UTILISED?

There is a lack of appropriately labelled datasets in PDD, making SSL a potential solution. SSL can perform classification and detection without relying on labelled datasets, thereby greatly reducing the model's dependence on such data. The application of SSL in PDD can be categorised into four primary categories: Generative, Predictive, Contrastive, and Hybrid. *Generative* methods acquire representations by generating or reconstructing input data or data akin to the input [23], [51]. *Predictive* methods establish a relevant pretext task suited to the dataset, generate self-generated labels, and train a model to predict these labels while learning data representations [4], [113]. *Contrastive* learning strategies involve training a model to discern between similar and dissimilar pairs of data points, aiming to construct a representation where similar points are clustered closely while dissimilar points are situated farther apart [103], [127]. *Hybrid* learning models amalgamate either generative,

predictive, or contrastive approaches with other neural network architectures [134], [135].

a: GENERATIVE MODELS

There are four generative methods proposed for PDD so far [23], [51], [104], [140]. An overview of the generative techniques employed in PDD is shown in Table 2.

Kim et al. [51] proposed Instance-Aware PDD (IA-PDD), a method leveraging spatial information through the integration of saliency maps and variational autoencoders. By employing combined vectors as conditional vectors of normalising flows, this approach achieved instance-awareness within the flow model. Testing on the PlantVillage and BRACOL datasets yielded AUROC scores of 0.98 and 0.99, respectively. However, IA-PDD is limited to single-class detection and encounters difficulties such as complex saliency mappings and high computational demands, raising concerns about its adaptability to varying environmental conditions.

In a similar vein, Yisen et al. [140] introduced the Autoencoder Self-supervised Learning Classifier (SSC-AE) for anomaly detection in strawberry samples using hyperspectral images. Initial curation of a manual dataset comprising bruise, fungal infection, and soil contamination categories led to the synthesis of anomalous spectra. The SSC-AE obtained an F1-score of 84.00% and an AUC of 90.8 on their dataset. However, the labour-intensive creation of hyperspectral images and the model's emphasis on low-dimensional data raise concerns about inefficiency and overfitting, limiting its usefulness in diverse datasets.

Furthermore, Liu et al. [104] proposed a pre-trained approach for classifying plant diseases and pests, focusing on Feature Relationship Conditional Filtering (FRCF) and Latent Semantic Masking Auto-Encoder (LSMAE). FRCF utilised a k-NN graph to select relevant data from the source domain, enhancing feature acquisition in the target domain. LSMAE improved feature representation discriminative power. The method obtained 99.93%, 74.79%, and 76.99% accuracy and 97.31%, 74.36%, and 75.02% F1-score on the PlantVillage, IP102, and CPB, respectively. Liu et al. [104] contradict SSL principles by utilising extensive datasets, raising concerns regarding overfitting and computational efficiency.

Jin et al. [23] introduced the Cascade Wavelet Attention Network (CWAN), a framework designed for synthesising and segmenting grape disease images. Prior to deploying CWAN, an unpaired PlantVillage dataset is synthesised through a meticulous blending process. CWAN extracts high-level features from the synthesised images while leveraging an autoencoder to capture low-level and colour features. Although CWAN achieved a PSNR of 47.854 and SSIM of 0.959 on the PlantVillage dataset, it demands significant computational resources due to extensive pre-processing. Furthermore, it faces challenges in accurately identifying the specific locations of disease areas and lacks a comparative analysis with other datasets.

TABLE 2. Overview of the generative methods used in plant disease detection.

Author	Method	Dataset	Accuracy	F1-Score	Performance AUC	AUROC	PSNR	SSIM
Kim et al. [51]	VAE	Plant Village, BRACOL	-	-	-	0.98, 0.99	-	-
Jin et al. [23]	CWAN	Plant Village	-	-	-	-	47.854	0.959
Liu et al. [104]	LSMAE	Plant Village, IP 102, CPB	99.93, 74.79, 76.99	97.31, 74.36, 75.02	-	-	-	-
Yisen et al. [140]	SSC-AE	Own	-	84.00	90.8	-	-	-

NOTE: CWAN = Cascade Wavelet Attention Network; LSMAE = Latent Semantic Masking Auto-Encoder; SSC-AE = Autoencoder Self-supervised Learning Classifier

b: PREDICTIVE MODELS

Yang et al. [4], [113] introduced the Location Feedback Classification Network (LFC-Net) approach in two different manuscripts for tomato and strawberry disease detection. The location network captures spatial information, while the feedback network’s guidance optimises iteration. Lastly, the classification network classifies tomato and strawberry diseases, utilising information from the location network. They achieved 99.7% accuracy in tomato disease detection using the PlantVillage dataset. For strawberry disease detection, they used their dataset, obtaining 92.48% accuracy, 88.45% F1-score, 90.68% precision, and 86.32% recall. However, the approach’s efficiency depends heavily on the image size parameter, K . Increasing the image size leads to higher feature dimensions, consequently increasing efficiency and computational complexity. To maintain efficiency, the image size is kept fixed at $K = 3$ for the entire model. Table 3 summarises the predictive methods utilised in PDD.

c: CONTRASTIVE MODELS

So far, negative sampling and clustering contrastive learning approaches have been used in PDD. Negative sampling involves pairing diseased plant images with healthy ones or images of plants with different diseases to learn distinguishing features. Conversely, clustering groups similar plant images to identify patterns and commonalities associated with specific diseases, thereby enhancing the model’s classification accuracy. Table 4 presents a concise overview of the contrastive methods employed in PDD.

I. **Negative sampling:** Zhang et al. [141] used a self-supervised Contrastive learning method (SimCLR) to detect cassava leaf diseases based on visual representations. The manuscript focused on comparing the effect of changing the data proportion of SimCLR in the classification task and achieved 91.59% accuracy on their dataset. However, the model’s requirements heavily depend on large batch sizes and data augmentation techniques. Moreover, focusing solely on changing the data ratio to improve accuracy may lead to overfitting and inefficiency in the other datasets. Similarly, Bunyang et al. [128] applied a SimCLR model as a pre-trained model to classify plant diseases, achieving 82.48% accuracy using their dataset. Nevertheless, this approach requires higher memory capacity and a longer computational time.

Zhang et al. [127], Yu et al. [103], and Meng et al. [142] have independently proposed the use of ViT for PDD. Zhang et al. [127] employed the entropy mask for classifying tea disease, extracting entropy information from the input dataset and fine-tuning it concurrently with ViT. Their method achieved 93.78% accuracy, 93.64% F1-score, 93.67% precision, and 93.80% recall on their tea disease dataset. Yu et al. [103] introduced a ViT method focusing on mixed attention for ultra-fine-grained categorisation tasks, achieving 91.0% accuracy. Meng et al. [142] used a pre-trained ViT model for recognising plant diseases and species, including known and unknown classes. Pre-trained on the PlantCLEF2022 dataset, their approach was tested on the PaddyDoctor and Own datasets to score 91.69 and 92.02 AUROC, respectively. However, the utilisation of a ViT necessitates a higher parameter count, leading to computational complexity. Zhang et al. [127] addressed this by sampling 25% of the original image, potentially losing crucial information for tea disease classification. Yu et al. [103] experienced suboptimal performance in accurately predicting disease regions, especially for apple foliar and SoyAgeing datasets, due to a high parameter count (35.42 million). Similarly, Meng et al. [142] faced higher computational complexity and reduced performance for similar plant diseases, highlighting the requirement for larger datasets.

II. **Clustering:** Monowar et al. [45] introduced an image clustering method for leaf disease identification, generating clusterable embeddings for robust disease classification. Their approach includes a novel mix-link and strong can-link pair strategy to enhance stability during training without labelled data, achieving 88.9% and 89.3% accuracy on PlantVillage and Citrus Disease datasets, respectively. Fang et al. [122] introduced the Cross-Iterative Kernel-based Image Clustering System (CIKICS) model employing a cross-iterative clustering process for PDD. The method encompasses four sequential stages: feature extraction, k-means clustering, image classification, and similarity management. The approach slightly improved the previous model, as it attained 89.1% accuracy on the same dataset. While Monowar et al. [45] claim efficiency across various datasets, reliance on data augmentation and random selection during the clustering may reduce

TABLE 3. Overview of the predictive methods in plant disease detection.

Author	Method	Dataset	Performance			
			Accuracy	F1-Score	Precision	Recall
Yang et al. [4]	LFC-Net	Plant Village	99.7	-	-	-
Yang et al. [113]	LFC-Net	Own	92.48	88.45	90.68	86.32

NOTE: LFC-Net = Location Feedback Classification Network

TABLE 4. Overview of the contrastive methods in plant disease detection.

Author	Method	Dataset	Performance				
			Accuracy	F1-Score	Precision	Recall	AUROC
Zhang et al. [141]	SimCLR	Own	91.59	-	-	-	-
Bunyang et al. [128]	SimCLR	Own	82.48	-	-	-	-
Zhang et al. [127]	ViT	Own	93.78	93.64	93.67	93.80	-
Yu et al. [103]	ViT	Own	91.0	-	-	-	-
Meng et al. [142]	ViT	Paddy Doctor, Own	-	-	-	-	91.69, 92.02
Monowar et al. [45]	Clustering	PlantVillage, Citrus Disease	88.9, 89.3	-	-	-	-
Fang et al. [122]	CIKICS	PlantVillage, Citrus Disease	89.1	-	-	-	-
Zhao et al. [53]	CLA	PlantVillage	90.52	-	-	-	-

NOTE: SimCLR = Self-supervised Contrastive Learning; ViT = Vision Transformer; CIKICS = Cross-Iterative Kernel-based Image Clustering System; CLA = Contrastive Learning for Leaf Disease Identification

effectiveness in diverse data scenarios, escalating computational complexity. Fang et al. [122] rely on pre and post-processing, increasing computational complexity and limiting usefulness to offline procedures and large datasets. Lastly, Zhao et al. [53] proposed the Contrastive Learning for Leaf Disease Identification (CLA) method, focusing on limited labelled and noisy unlabelled training data. CLA includes pre-training with large-scale unlabelled data and fine-tuning with domain adaptation. The domain adaptation layer (DAL) within CLA aids in aligning labelled and unlabelled data, thereby generating more generalised visual representations with a mix-ratio adjustment in the source domain. The CLA method achieved 90.52% accuracy on PlantVillage. However, the mix ratio in the source domain may impact CLA's accuracy, depending on the domain's size and class count.

d: HYBRID MODELS

Bedi et al. [134] introduced a hybrid self-supervised model for peach PDD, combining a Convolutional Autoencoder (CAE) and Convolutional Neural Network (CNN) to achieve high accuracy with fewer parameters. The CAE downstream features from the PlantVillage dataset, while the CNN classifies these features, achieving 98.38% accuracy. However, Bedi et al. [134] only focused on peach plant diseases, lacking cross-validation with other datasets, raising concerns about generalisability to other plant diseases. Although it utilises fewer parameters, the method's accuracy is lower compared to other methods, and overfitting problems may arise after 18 epochs.

Cao et al. [135] proposed a hybrid SSL approach for cucumber PDD, comprising three branches: multi-modal contrastive, self-supervised contrastive, and label information module. These branches map semantic information, create distance between two enhancement images, and generate

uniform spaces for categories, respectively. Achieving 90.63% accuracy, 79.92% F1-score, 92.85% precision and 77.28% recall in their own dataset, Cao et al. [135] demonstrate promising results. However, the method's reliance on combining multiple learning methods simultaneously increases computational complexity, and a lack of comparison with other SSL methods raises questions about its effectiveness.

A summary of the two hybrid methods is shown in Table 5.

2) RQ2: WHICH DATASETS HAVE BEEN EMPLOYED?

Like the application of ML in many other areas, datasets play an important role in PDD so that ML algorithms can provide expected performance. Researchers have used various datasets, some of which are publicly available and some of which are self-generated and not publicly available. Table 6 provides a summary of datasets used in PDD with SSL methods. In addition, a few sample images of different public datasets are also shown in Figure 8.

- I. **PlantVillage:** The PlantVillage dataset is among the most widely used publicly available datasets. Introduced by Penn State University, it comprises a multi-plant dataset designed for disease categorisation. This dataset contains around 54,309 images of plants spanning 14 distinct species. The images are categorised into multiple disease types, including 4 bacterial diseases, 17 fungal diseases, 1 mite disease, and 2 mould diseases. Furthermore, the dataset includes images of 12 different types of healthy plant species unaffected by diseases [143].
- II. **Citrus Disease:** The University of Gujrat introduced the citrus disease dataset, which consists of approximately 759 images of citrus fruits and leaves, both healthy and diseased. These images were captured in Pakistan using a DSLR camera under the guidance of subject matter experts. The dataset includes

TABLE 5. Overview of the hybrid methods in plant disease detection.

Author	Method	Dataset	Performance			
			Accuracy	F1-Score	Precision	Recall
Bedi et al. [134]	CAE and CNN	PlantVillage	98.38	-	-	-
Cao et al. [135]	VLP and CLIP	Own	90.63	79.92	2.85	77.28

NOTE: CAE = Convolutional Autoencoder; CNN = Convolutional Neural Network; VLP = Vision-Language Pre-training; CLIP = Contrastive Language-Image Pre-training

TABLE 6. Datasets used for self-supervised models in plant disease detection.

Authors	Dataset								
	PV	CD	Br	PDoc	IP102	CPB	PaDoc	CDS	Others
Monowar et al. [45]	✓	✓							
Kim et al. [51]	✓		✓						
Fang et al. [122]	✓	✓							
Zhao et al. [53]	✓			✓					
Jin et al. [23]	✓								
Liu et al. [104]	✓				✓	✓		✓	
Yang et al. [4]	✓								
Bedi et al. [134]	✓								
Yu et al. [103]									✓
Meng et al. [142]							✓		✓
Yang et al. [113]									✓
Zhang et al. [141]									✓
Yisen et al. [140]									✓
Cao et al. [135]									✓
Zhang et al. [127]									✓
Bunyang et al. [128]									✓
Purpose dataset	Cla	Cla	Seg & Cla	Cla	Det	Cla	Cla	Cla	NA

NOTE: PV = PlantVillage; CD = Citrus Disease; Br = BRACOL; PDoc = Plant Doc; CPB = Citrus Pest Benchmark; PaDoc = Paddy Doctor; CDS = Corn Disease and Severity Dataset; Cla = Classification; Seg = Segmentation; Det = Detection.

approximately 609 pictures of citrus leaves, categorised as healthy, canker, melanosis, black spot, and greening. Additionally, there are nearly 150 categorised photos of citrus fruits [144].

- III. **BRACOL:** The BRACOL dataset is a unique collection designed to improve the diagnosis of diseases in Arabica coffee leaves. It consists of 1,747 images, divided into two sets: 500 labelled images and 1,247 unlabelled images. The labelled set includes images of both healthy and diseased leaves, categorised into background, leaf, and symptom classes. The symptom class further delineates five distinct disease appearances. Initially captured at high-resolution images (1024×2048 pixels), the images are resized to 256×256 pixels using bilinear interpolation to match the model input specifications. The dataset stands out for its precise pixel-level annotation for accurate segmentation masks and classification and segmentation models. Additionally, its extensive collection of unlabelled images serves as a foundation for pre-training feature extractors, enhancing models' ability to generalise and recognise plant diseases [145].
- IV. **PlantDoc:** The PlantDoc collection consists of 2,598 RGB images depicting leaf diseases across 13 distinct plants. These images are categorised into 27 disease types and were captured directly from their natural

habitats. This real-world sourcing of data enhances the training of ML models by providing a diverse array of intricate information for the adaptability of PDD technologies [146].

- V. **IP102:** The IP102 dataset contains 75,222 images representing 102 insect pest classes. Over 19,000 images are tagged with bounding boxes, enabling the development of object detection algorithms. The dataset's hierarchical structure categorises pests based on their impact on specific plants, facilitating the development of models for efficient pest management [147].
- VI. **Citrus Pest Benchmark (CPB):** Developed by Bollis et al. in 2020 [148], the CPB dataset comprises 10,816 high-resolution images ($1,200 \times 1,200$ pixels) classified into two categories: mite-infected and negative. The mite-infected category is further subdivided into six particular varieties of mites. These images were taken with a Samsung Galaxy A5 smartphone equipped with a 13 MP camera and a $60\times$ magnifier. The smartphone is equipped with white and ultraviolet LED lights to aid image capture. The images were then downsized to 224×224 pixels for ML purposes.
- VII. **PaddyDoctor:** The Paddy Doctor dataset comprises 16,225 high-resolution photographs ($1,080 \times 1,440$ pixels) of paddy leaves categorised into 13 categories. These categories include 12 diseases as well as

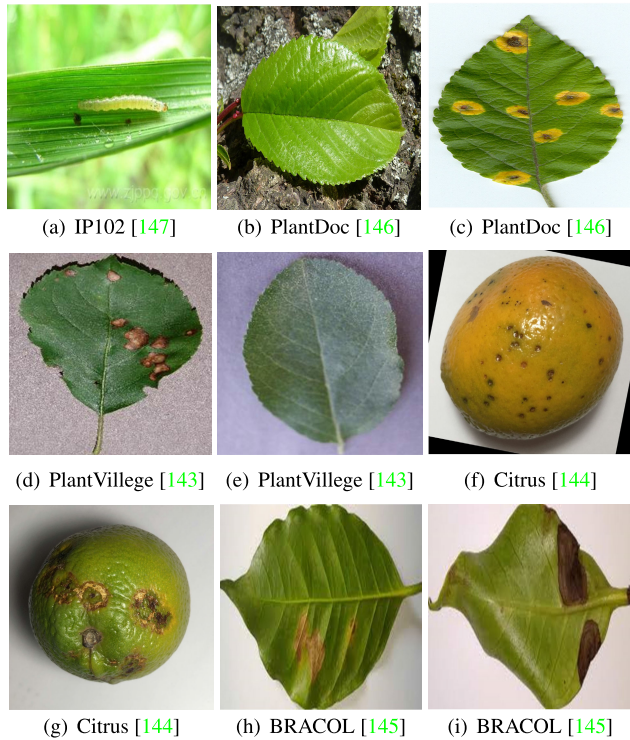


FIGURE 8. Sample images of different public datasets where (a) is used for detection, (b)-(g) are used for classification, and (h) and (i) are used for both classification and segmentation.

healthy leaves. This extensive dataset facilitates the development of ML models for disease diagnosis, promoting progress in paddy disease management and facilitating sustainable farming practices through early detection and treatment of plant diseases [149].

VIII. Corn Disease and Severity Dataset: The corn disease and severity dataset consists of 1,597 images captured in the field settings, which are categorised into three main types of corn diseases: Gray Leaf Spot (GLS), Northern Leaf Blight (NLB) and Northern Leaf Spot (NLS). These images were collected using handheld sensors and were further enhanced with augmented backdrops to accurately represent the intricate and diverse conditions present in natural habitats. This dataset offers a strong basis for creating machine-learning models capable of diagnosing and evaluating the severity of corn illnesses. With a wide variety of images of each disease category, including 524 for GLS, 511 for NLB, and 562 for NLS [150].

Other self-supervised approaches, such as [113], [135], and [141], used customised datasets for PDD. Besides, the Ultra-fine-grained dataset was also used [103] for the ultra-fine-grained categorisation task. Additionally, several public datasets, including Embrapa [61], AES-CD9214 [151], and LWDCD2020 [89], are available for PDD but have not been utilised by self-supervised models. Researchers have focused on different plants and parts of plants for PDD

using SSL models. Table 7 shows that most researchers focused on plant leaves and considered single or multiple plant species. For instance, Jin et al. [23] focused on grape leaves, while Fang et al. [122] investigated a broader range of plants, including corn, grapes, apples, peaches, peppers, tomatoes, strawberries, potatoes and cherries for PDD using SSL techniques.

3) RQ3: WHICH PERFORMANCE METRICS HAVE BEEN USED TO EVALUATE THE EFFECTIVENESS OF THE MODELS?

Evaluating every ML algorithm to identify its effectiveness and accuracy in any particular application is very important. Several performance parameters have been formulated to assess the efficacy of SSL techniques for PDD. Accuracy, F1-score, AUC (Area Under the Curve), AUROC (Area Under the Receiver Operating Characteristic curve), precision, PSNR (Peak Signal-to-Noise Ratio), SSIM (Structural Similarity Index Measure) and recall are commonly used performance parameters in SSL for PDD.

- I. **Accuracy:** Accuracy in ML refers to the proportion of correct predictions a classification model makes. It is a basic metric used to assess the overall efficiency of a model in accurately categorising the data and is formulated as

$$\text{Accuracy} = \frac{|TP| + |TN|}{|TP| + |TN| + |FP| + |FN|}, \quad (1)$$

where TP, TN, FP, and FN denote sets of true positive, true negative, false positive and false positive cases, respectively, and $|\cdot|$ denotes set cardinality.

- II. **Precision (P):** P is the ratio of correctly predicted positive observations to the total predicted positives, which is derived as Equation 2 [42].

$$P = \frac{|TP|}{|TP| + |FP|}. \quad (2)$$

- III. **Recall (R):** R is the ratio of correctly predicted positive observations to all observations in the actual class, which is derived as Equation 3 [41].

$$R = \frac{|TP|}{|TP| + |FN|}. \quad (3)$$

- IV. **F1-score:** The F1-score is a performance measure utilised in ML to amalgamate precision and recall into a singular measure, which is executed as equation 4. The F1-score is calculated as the harmonic mean of precision and recall. It offers a more thorough assessment of a model's performance by taking into account both FP and FN, which are both important in the context of the application [104].

$$\text{F1-score} = 2 \times \frac{P \times R}{P + R}. \quad (4)$$

- V. **AUC:** AUC is a performance statistic commonly employed in ML to assess the efficacy of binary classification models. The AUC is a metric that quantifies the area under the Receiver Operating

TABLE 7. Different kinds of plants and plant parts for self-supervised models in plant disease detection.

Authors	Plants	Plants Parts
Monowar et al. [45]	Apple, corn, grape, peach, pepper, potato, and tomato	leaf
Kim et al. [51]	Arabica coffee	Leaf
Fang et al. [122]	Apple, corn, grape, peach, pepper, potato, tomato, strawberries, and cherry	Leaf
Zhao et al. [53]	Apple, corn, grape, peach, pepper, potato, tomato, strawberries, and cherry	Leaf
Jin et al. [23]	Grape	Leaf
Liu et al. [104]	Corn	Leaf
Yang et al. [4]	Tomato	Leaf
Yu et al. [103]	Cotton, soy, and apple	Leaf
Meng et al. [142]	Rice, tomato, rose, and cotton	leaves, stems, and heads
Yang et al. [113]	Strawberry	15 diseases in different parts
Zhang et al. [141]	Wheat	Leaf
Liu et al. [140]	Strawberry	Leaf
Bedi et al. [134]	Not specific	Leaf (bacteria spot)
Cao et al. [135]	Cucumber	Leaf
Zhang et al. [127]	Tea	Leaf
Bunyang et al. [128]	Not specific	Leaf

Characteristic (ROC) curve. This graph represents the relationship between the True Positive Rate (TPR) and the False Positive Rate (FPR) at different threshold values. The AUC value is a concise metric that captures the overall performance of a model across various classification thresholds. It quantifies the model's capacity to differentiate between the two classes. Equation 5 provides the mathematical derivation of the AUC from the ROC curve [140].

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(x)) dx, \quad (5)$$

where

$$\text{TPR} = \frac{|\text{TP}|}{|\text{TP}| + |\text{FN}|} \text{ and } \text{FPR} = \frac{|\text{FP}|}{|\text{FP}| + |\text{TN}|}.$$

VI. AUROC: AUROC is a metric used in ML to evaluate the performance of classification models across different threshold settings. The ROC curve is a visual representation that demonstrates the diagnostic performance of a binary classifier system as the discrimination threshold is varied. The AUC quantifies the whole area underneath the complete ROC curve, ranging from the point (0,0) to (1,1). The AUROC value varies between 0 and 1 [51].

VII. PSNR: PSNR is an often employed measure in ML, particularly for tasks related to image processing like image reconstruction, compression, and super-resolution. It quantifies the quality of a reconstructed or compressed image relative to the original image. Equation 6 expresses the PSNR in the logarithmic

decibel scale [23].

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{\text{Max}_i^2}{\text{MSE}} \right), \quad (6)$$

where Max is the highest possible pixel value that can be assigned to an image, and MSE (Mean Squared Error) quantifies the difference between the original image and the reconstructed image.

VIII. SSIM: SSIM is utilised to evaluate the similarity and distortion between two images. SSIM is derived from the calculation of three components that evaluate the brightness, difference in intensity, and structure of a reference image and a test image. The SSIM metric has a numerical range from -1 to 1, with a value of 1 indicating complete similarities between two images [23].

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)}, \quad (7)$$

where

- x and y are two images;
- μ_x and μ_y are the average values of x and y , respectively;
- σ_x^2 and σ_y^2 are the variances of x and y , respectively;
- σ_{xy} is the covariance of x and y ; and
- c_1 and c_2 are constants to stabilise the division with a weak denominator.

B. DISCUSSION

SSL approaches enable models to gain an implicit understanding of the structure and characteristics of the data, enabling them to perform tasks like classification and detection without relying heavily on a labelled dataset. These methods play an important role in the PDD as there is often a scarcity of annotated datasets. Hence, researchers are

motivated to apply SSL approaches to detect plant diseases discussed in section. Though these approaches have made tremendous contributions, some research challenges remain in using SSL for PDD. The primary research challenges are generalisability, reliance on extensive datasets, computational complexity, and lack of validation, which are discussed below with future directions.

1) GENERALISABILITY

The importance of generalisability in PDD lies in its capacity to enable models to properly identify and diagnose diseases in many plant species and under varying environmental and geographic conditions without the need for frequent updates. However, generalisability is one of the research challenges for SSL in PDD. Lack of data diversity and dependency on pre-trained models are the primary constraints for PDD's generalisability using SSL approaches.

a: LACK OF DATA DIVERSITY

There are many types of plant species and their various diseases available in the agricultural field. Besides, there is also a possibility of new diseases and their mutant in the passage of time. These diseases also have different symptoms and characteristics in different plants, which cannot always be mapped to a common feature space. The SSL models are employed to detect specific plant diseases, which limits their ability to generalise across different plant species and disease types. As the SSL models [23], [134] are trained with a particular dataset, detecting all these species and their diseases is very challenging. For example, the PlantVillage dataset contains images of only 14 plant leaf species with fixed backgrounds in laboratory conditions [143]. Hence, It will be tough for an SSL model that is trained by the PlantVillage dataset to detect all these species and their diseases. While some researchers have customised their own datasets, issues of overfitting and validation remain a concern. In addition, environmental factors such as weather, light, and soil type also influence disease conditions, complicating the development of a universally applicable model. Hence, the lack of diverse datasets exacerbates this issue, preventing models from being trained in varied conditions and leading to poor generalisation. To address this, there is a need to develop a benchmark dataset for training an SSL model that contains images of diverse plant diseases with diverse conditions, such as various plant species, disease complexities, variable lighting, weather and background.

b: DEPENDENCY ON PRE-TRAINED MODEL

There are some SSL models [104], [128] that utilise pre-trained models for the pretext task, leading to the limitation of generalisability. These pre-trained models were trained on large datasets like COCO and ImageNET, which are not under the domain of plant diseases. Therefore, it is very challenging to learn distinct features of different plant diseases and may also have negative learning, as mentioned

in [152]. Fine-tuning a pre-trained model on a particular dataset can reduce the model's diversity and flexibility. Pre-trained models that were optimised for specific tasks may not perform well on different tasks, leading to issues with explainability and transparency. This can result in reduced compatibility, generalisability, and efficiency when applied to unknown and unseen datasets. It is also mentioned in different research, like [153], [154], that pre-trained models are trained with balanced and organised datasets, which may have poorer accuracy due to the overfitting caused by new unbalanced and unorganised datasets. To address this, the models should be trained on the specific plant disease dataset to use in PDD tasks.

2) RELIANCE ON EXTENSIVE DATASETS

In the agricultural field, especially in PDD, labelled datasets are rare, expensive, and time-consuming and require skilled workers and proper infrastructure to generate those datasets. That is why SSL methods are recently being proposed in PDD to avoid using these large labelled datasets. However, some of the proposed SSL methods like [142] require larger datasets to train the models for achieving proper accuracy. Even some of the SSL approaches utilise pre-trained models [104], [128], which are also trained on larger labelled datasets, as discussed previously. The necessity of these larger datasets is against the major objective of the SSL approaches. Therefore, the SSL method needs to be trained efficiently in such a way that it can provide high accuracy using a smaller dataset.

3) COMPUTATIONAL COMPLEXITY

For early PDD, computational simplicity is very important. Many SSL methods [4], [113], [142] employed in PDD involve complex architectures with millions of parameters, which increase the computational complexity of the entire system. Additionally, these models often rely on extensive preprocessing and postprocessing approaches [23], [122], further adding to the computational burden. Besides, there are some pre-trained models [104], [128] used in SSL for PDD, which are trained with larger image sizes and higher parameters. This can lead to higher computational complexity. In that case, an SSL model ought to be designed with reduced computational complexity and fewer parameters. Additionally, it should utilise lower, compatible image sizes and a more simplified structure to detect plant diseases.

4) LACK OF VALIDATION

A proper validation process is a must for having a reliable proposed model. The SSL approaches that are used in the particular task of PDD should be compared with other existing SSL methods to validate the improvement and efficiency. However, most of the SSL methods [23], [103], [122] applied in the PDD were not compared with existing SSL methods. These methods are mostly compared with either supervised machine learning or deep learning models to validate the performances, which leads to unparalleled

comparison. Thus, the performance of the developed SSL models should be compared with other existing SSL models to have proper validation and comparison.

V. CONCLUSION

Early PDD has a vital role in agriculture, enriching plants' quality, quantity, and productivity. A systematic review has been conducted on PDD using SSL models to know how plant disease can be identified in the scarcity of annotated datasets. The self-supervised models used in PDD were discussed in four categories: generative, predictive, contrastive and hybrid. It can be concluded that, though researchers are using SSL models in PDD due to their scalability and higher efficiency with unlabelled datasets, they need to face certain challenges. The challenges of self-supervised models on PDD include depending on the generalisability, reliance on extensive datasets, computational complexity and lack of validation. The challenges show the opportunity for the researchers to improve the SSL models in PDD in terms of generalisability, lower computational complexity and less dependency on a particular dataset. A diverse public dataset will also be a key factor for the efficiency of this research.

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